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Lesions Distribution, Nature and Severity Along the Gastrointestinal Tract in Cats With Low-Grade Intestinal T-Cell Lymphoma or Lymphoplasmacytic Enteritis

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ABSTRACT

Background: The distribution and severity of lesions in low-grade intestinal T-cell lymphoma (LGITL) and lymphoplasmacytic enteritis (LPE) might vary within the gastrointestinal (GI) tract and coexist.

Objectives: Evaluate the distribution of the lesions in cats diagnosed with LPE or LGITL and assess the presence of concurrent inflammatory infiltrates and lesion severity in LGITL cases.

Animals: Fifteen cats diagnosed with LGITL and 12 cats with LPE.

Methods: Prospective descriptive study including cats diagnosed with LGITL or LPE based on histology and immunohistochemistry between 2019 and 2022. Endoscopic and surgical biopsies of all gastrointestinal segments were performed for histopathologic evaluation. For each LGITL case, concurrent inflammatory infiltrates and the severity of histologic lesions were assessed.

Results: Jejunum was systematically affected, followed by ileum (LGITL: 93% [14/15]; LPE: 92% [11/12]) and duodenum (LGITL: 87% [13/15]; LPE: 100% [12/12]). Stomach (LGITL: 29% [4/14 cats]; LPE: 33% [4/12]) and colonic (LGITL: 2/3; LPE: 3/4) infiltration were less common. Gastroduodenoscopy correctly diagnosed 87% of LGITL cats, but combined upper and lower GI sampling was diagnostic for all LGITL cases. In LGITL cases, stomach exhibited less severe lesions than duodenum, jejunum, and ileum ($p < 0.001$, respectively). In 9/15 LGITL cats (60%), concomitant inflammation was found in segments infiltrated with neoplastic lymphocytes.

Conclusion and Clinical Relevance: LGITL shows a heterogeneous distribution within the digestive tract. Endoscopy sampling of the upper and lower GI tract can be considered a first-line diagnosis method. Signs of concurrent inflammation were found in LGITL cases, supporting a possible *continuum* between LGITL and LPE.

1 | Introduction

Chronic enteropathies, including low-grade intestinal T-cell lymphoma (LGITL) and lymphoplasmacytic enteritis (LPE), are increasingly common in elderly cats [1–4]. Differentiating LGITL from LPE can be challenging as clinical signs and features of non-invasive diagnostic work-up overlap [1, 2, 5]. Therefore, histopathologic and immunohistochemical evaluation of intestinal biopsies remains the gold standard to differentiate those two diseases and is essential as prognosis and treatment differ [2, 6].

Abbreviations: GI, gastrointestinal; LGITL, low-grade intestinal T-cell lymphoma; LPE, lymphoplasmacytic enteritis; US, ultrasound.

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Concurrent inflammatory lesions have been reported in up to 60% of LGITL cases and might be due to a *continuum* between these two entities, which further complicates their distinction and highlights the necessity of immunohistochemistry for ambiguous cases [2, 6–10]. Yet, the best method for digestive tract biopsies remains unknown, and the distribution of inflammatory or neoplastic lesions can vary widely within the gastrointestinal (GI) tract [2]. The LGITL involves at least one segment of the digestive tract, with the jejunum almost always affected, followed by ileum, duodenum, less frequently stomach, and rarely colon [2, 10–13]. However, most of those studies evaluating the distribution of lesions use different histologic, immunohistochemical, and clonality criteria than recently described and validated [2].

Endoscopy is a quick and minimally invasive procedure allowing clinicians to perform multiple mucosal biopsies of the stomach, duodenum, colon, and, with appropriate endoscopic equipment, distal ileum and proximal jejunum [2, 14]. However, this procedure has some drawbacks. First, the middle-to-distal jejunum is often not reached despite being commonly affected by LGITL. Second, to the authors knowledge, LGITL is not characterized by any specific macroscopic lesion [2, 14–18]. Additionally, in some cases, the accessibility of some GI segments can be limited by pyloric narrowing or difficulties in ileal intubation [19]. Surgery can be elected as it allows sampling of all abdominal organs, including the whole jejunum [1–3, 6]. However, it might be contraindicated in some elderly cats with multiple comorbidities, life-threatening complications can occur, and it requires treatment postponement until wound healing is complete [20, 21]. Comparative diagnostic performance between surgical and endoscopic biopsy sampling in this clinical context is lacking in the literature. Some studies suggest that endoscopically collected biopsy specimens were inadequate for accurate differentiation between LPE and LGITL [22]. However, a lower number of biopsies were performed in this study, and the adequacy of some specimens was questioned by pathologists. Finally, a recent study confirms a systematic apical-to-basal neoplastic infiltration emerging from the mucosa in 100% of cases [6], which suggests that endoscopic mucosal biopsies are adequate for diagnosing LGITL.

Therefore, the aims of this study were: (1) to describe the distribution of lesions within the GI tract in cats diagnosed with LPE and LGITL and analyze biopsies performed both by endoscopy and surgery during the same procedure, (2) to evaluate the presence of concurrent inflammatory lesions in LGITL cases, and (3) to assess the severity of the lesions within the different GI segments in LGITL cases. A secondary aim of this study was to compare epidemiologic features, clinical signs, and selected laboratory and imaging findings in cats diagnosed with LGITL and LPE.

2 | Materials and Methods

2.1 | Study Design and Inclusion Criteria

This prospective convenience sample study was conducted in one referral center (École Nationale Vétérinaire d'Alfort, Paris, France) between 2019 and 2022. Data were prospectively collected, except for some biochemical, hematological, and abdominal ultrasound (US) data obtained shortly before inclusion, which were retrospectively reviewed. Ethical approval was validated by the ethical committee of our institution (ENVA COMERC no 2017-05-09). All owners gave written consent for all investigations performed.

Cats were eligible for enrollment in this study if they displayed signs of GI disease (weight loss, vomiting, diarrhea, lethargy, anorexia) for more than 3 weeks duration, that did not improve after symptomatic treatment (gastro-intestinal protectant, antibiotics, acid suppressant therapy), or dietary modification after exclusion of infectious, extra-digestive, and metabolic disorders. If the cat had received glucocorticoid treatment at the time of inclusion, a progressive washout was attempted before intestinal biopsies were performed for all cases (2 weeks were recommended). Eligible cats were then included if they had a histological and immunohistochemical diagnosis of LGITL or LPE based on recent diagnostic criteria [2].

Cats were excluded if they had untreated comorbidities that could explain the clinical signs (e.g., hyperthyroidism), a medical condition contraindicating general anesthesia or laparotomy (e.g., congestive heart failure, severe hypoalbuminemia, advanced chronic kidney disease), or another neoplasia.

2.2 | Clinical, Laboratory and Ultrasonography Information

Clinical data including age, sex, weight, breed, nature, and duration of clinical signs, physical examination, diet, and treatment history were recorded.

Complete blood cell count (CBC), serum biochemistry panel, fecal analysis, serum total thyroxine (T4) concentration, feline pancreas-specific lipase (f-PL), and serum cobalamin (vitamin B12) concentration were obtained in most cats either by IDEXX laboratories, France, or by BioPôle laboratory at École Nationale Vétérinaire d'Alfort, Paris, France. The reference intervals for the variables of both laboratories are provided in Table S1. Some laboratory data were also provided by the referring veterinarian if they had been performed within the past month. Hypocobalaminemia was defined by a serum cobalamin concentration lower than the reference interval of the laboratory used [2, 23].

Abdominal US was performed at the same referral center for all cats, except for one, for which it was performed by an experienced veterinarian in imaging diagnosis 2 weeks prior to referral. All examinations carried out at the referral center were performed by a board-certified radiologist or a resident under supervision. Standardized images were collected, including all regions of the GI tract (stomach, duodenum, jejunum, ileum, caecum, and colon) and regional lymph nodes. For each GI segment, loss, alteration, and thickening of wall layering were evaluated and recorded as focal, multisegmented, or diffuse. Thickness, echogenicity (i.e., normal, hypoechoic, hyperechoic) and shape (i.e., normal, rounded, irregular) of jejunal lymph nodes were evaluated. The presence or absence of perinodal fat echogenicity (i.e., normal or hyperechoic) was also recorded.

Abdominal effusion and generalized steatitis also were recorded as either present or absent.

2.3 | Collection of Gastro-Intestinal Biopsy Material

For all cats, during the same standardized anesthetic procedure, upper GI endoscopic biopsies were performed, immediately followed by surgical biopsies of the jejunum. Ileal sampling was systematically performed, either through ileo-colonoscopy or during surgery. Cats for which the dual surgical and endoscopic procedure was declined (except one cat which had only surgical biopsies of the duodenum, jejunum and ileum) were excluded from the study.

Cats were positioned in left lateral recumbency as previously recommended, and a gastro-duodenoscopy was performed using a GIF 180 pediatric video gastroscope (Olympus, France), with an external diameter of 8.8 mm. A macroscopic evaluation of the stomach was first conducted, followed by an examination of the pyloro-duodenal junction and duodenum. Eight to 10 biopsies in each stomach and duodenum were collected using a reusable biopsy forceps with oval fenestrated jaws. If the pylorus was not successfully passed after 3 attempts, blind biopsies were performed through pyloric catheterization.

Concerning ileal sampling, if ileo-colonoscopy was performed, 4 to 6 biopsies were taken by blinded catheterization. If the ileocolic junction was unsuccessfully catheterized during the endoscopic procedure, surgical biopsies of the ileum were performed afterward. Colonic macroscopic evaluation and biopsies were not mandatory in this study and were performed based on the clinical signs, duration of the endoscopic procedure, and anesthesia concerns. When performed, 8–10 biopsies of the colon using a reusable biopsy forceps with oval fenestrated jaws were obtained. The overall duration of the endoscopic procedure was about 30–45 min.

For each segment endoscopically evaluated during this procedure, the presence or absence of macroscopic lesions was recorded according to the WSAVA endoscopic guidelines [14]. Lesions reported were erosions or ulcers, oedema, hyperemia, discoloration, friability of the mucosa, hemorrhage, and specifically lacteal dilation or excessive granularity for the duodenum evaluation. Severity of those macroscopic findings was graded from 1 to 3. Unsuccessful or difficult pylorus passage (graded 1 to 3/3) or ileum catheterization was recorded for each case.

Surgical full-thickness biopsies of the jejunum and in some cases distal ileum (i.e., 2–3 biopsies in total) were then performed. Sample sites were chosen according to sonographic features in addition to visual and tactile inspection of intestinal loops during celiotomy (induration and color change).

Cats were then hospitalized for 2–4 days after the surgery to monitor for postoperative complications. All animals underwent serial measurements of hematocrit and serum albumin concentration, and repeated abdominal ultrasonography to detect any surgical complications.

2.4 | Histology

All intestinal segment samples (e.g., stomach, duodenum, jejunum, ileum and colon when performed) were fixed separately in 10% neutral-buffered formalin and submitted to a veterinary pathology department (École Nationale Vétérinaire d'Alfort, BioPôle, Paris, France) for histologic evaluation. Specimens were embedded in paraffin and cut into 4 μm-thick sections. All tissue sections stained with hematoxylin & eosin (HE) were blindly reviewed by a board-certified veterinary pathologist. Even if multiple biopsies were performed in the same segment, the pathologist provided a single diagnosis (e.g., LGITL, LPE, no lesion) and a single severity score for each GI segment, as described below.

For each intestinal segment analyzed, the epithelium and lamina propria were reviewed separately and the segment sample was classified as LGITL, LPE, or subnormal according to histopathologic criteria previously reported [6]. If at least one segment was infiltrated by neoplastic lymphocytes, the cat was diagnosed with LGITL. Otherwise, if no GI segment showed neoplastic infiltration but inflammatory lesions compatible with LPE, the cat was diagnosed with LPE. If no lesions were identified, the mention "no lesion" was reported.

The subjective severity of histopathologic lesions was graded on a scale from 1 to 3 (e.g., mild, moderate, severe inflammatory or neoplastic infiltration, respectively) for each GI segment examined. When LGITL was diagnosed in an intestinal segment, the presence or absence of any concomitant inflammatory lesion was recorded (e.g., prominent or minor cellular polymorphisms with mixed lymphoid and plasmocytic inflammatory infiltrates, neutrophilic cryptitis, abscesses).

For ambiguous cases where neoplastic and inflammatory lesions could not be differentiated with histopathology alone, immunohistochemistry was reviewed to confirm the diagnosis, as described below. Clonality testing was not used for initial diagnosis in our study, as recent data have reported a high percentage of clonality in LPE cases, likely due to the strongly suspected *continuum* between the two diseases [2, 6].

For four cats, immunohistochemistry was performed in another anatomopathological laboratory and was unavailable for review at our institution. For the rest of the study cohort, sections from each sample were labeled using a panel of antibodies. Anti-CD3 antibody staining was performed to identify the T-cell lineage (1:20 dilution, F7.2.38, DakoCytomation, Santa Clara, California). Anti-CD20 antibody staining was used to identify the B-cell lineage (1:400 dilution, RB-9013-P1c, Lab Vision Corporation, Fremont, California). The MIB-1 antibodies were examined with respect to Ki67 proliferative index activity within the neoplastic population (1:75 dilution, MIB-1, DakoCytomation, Santa Clara, California). The Ki67 index is preferentially expressed during all active phases of the cell cycle, with peak expression during the M phase, and then disappears in resting cells. Evaluation of CD3 and CD20 staining was mandatory for this study on duodenum, jejunum, and ileum samples. Ki67 evaluation was performed at the discretion of the clinician and the feasibility of this analysis given the remaining samples available. Immunohistochemistry on

the stomach and colon was performed depending on the HE tissues sample evaluation and the histological suspicion of neoplastic infiltration. Quantitative assessment in percentage was performed on the lamina propria and epithelial compartment of each segment when immunohistochemistry was performed, and in the most infiltrated areas if infiltrates were heterogeneously distributed.

2.5 | Data Analysis and Statistics

All statistics were performed by using a computer software (Addinsoft XLSTAT: statistical and data analysis solution, version 2020). Continuous variables were assessed for normality by the Shapiro–Wilk test and histogram distribution. As the data were not normally distributed, quantitative data were reported as median (interquartile range (IQR); minimal and maximal values). Qualitative variables were reported as numbers and percentages (%) when relevant.

A Chi-squared test with Yates correction or Fisher's exact tests, when expected counts within cells were less than 5, was used for qualitative variables. The Kruskal Wallis test was performed for comparing quantitative data, using a Dunn-Bonferroni test for multiple comparisons. For all analyzes, the level of significance was set at $p < 0.05$.

3 | Results

3.1 | Study Cohort, History and Signalment

Fifty-six cats met the inclusion criteria during the study period. Two cats, later diagnosed with LGITL, were excluded due to concurrent digestive mastocytoma, and 27 cats were excluded as the dual surgical and endoscopic approach was declined by the owner. No cat was excluded from the study due to extra-intestinal diseases. Therefore, 27 cats were included in this study.

Fifteen cats were finally diagnosed with LGITL and 12 cats with LPE. The signalment of those cats is presented in Table 1. There was no significant statistical difference between the two groups.

TABLE 1 | Epidemiologic data and signalment of cats diagnosed with low-grade intestinal T-cell lymphoma (LGITL) and lymphoplasmacytic enteritis (LPE).

	LGITL ($n = 15$)	LPE ($n = 12$)	p[a]
Median age (year old)	12 (11.5–13.5; 8–15)	13.5 (12.5–14; 6–15)	0.294
Median weight (kg)	3.7 (3.26–4.65; 2.6–6)	4.075 (3.35–5.1; 1.7–9)	0.608
Sex distribution	10 female (67%)	5 female (42%)	0.363
	5 male (33%)	7 male (58%)	
Sprayed/neutered	13/15 (87%)	11/12 (92%)	1
Indoor life	13/14 (93%)	8/10 (80%)	0.55
Commercial diet[b]	8/14 (57%)	4/10 (40%)	0.679
Breed	Domestic shorthair: 12	Domestic shorthair: 7	0.661
	Siamese: 1	British shorthair: 1	
	Chartreux: 1	Oriental: 1	
	Persian: 1	Chartreux: 1	
		Somali: 1	
		Burmese: 1	

Note: Continuous data are expressed as median (IQR; range); categorical data are expressed as number (%).
[a]Reported p values are for the Mann–Whitney test (continuous data) or either $\chi 2$ or Fisher's exact test (where expected values were < 5 for at least 1 in the contingency table; proportional data).
[b]Other than commercial diet were hyperdigestible, hypoallergenic, anallergenic and home-made diet.

In the LGITL group, the previous use of corticosteroids within 3 months was known for 14 cats. Five out of 14 (33%) cats had received corticosteroids within 3 months prior to diagnosis at a median dosage of 1 mg/kg/d (0.8–1.7; 0.6–2). The median duration of treatment was 183 days (31–503; 25–730), and the median time between biopsy sampling and corticosteroid withdrawal was 10 days (9–15; 3–20). Concurrent diseases previously diagnosed in these 15 cats were hyperthyroidism (4/15), chronic kidney disease (5/15), pancreatitis (1/15) and hypercalcemia of unknown origin (1/15).

In the LPE group, 2/12 (17%) cats had received corticosteroid therapy within 3 months prior to the diagnosis at a median dosage of 0.8 mg/kg/d (0.4–1.1; 0.1–1.5). The duration of corticosteroid treatment was known for only one cat and was 59 days. The time between biopsy sampling and treatment end was 8 and 23 days for those two cats. Concurrent comorbidities diagnosed in those 12 cats were chronic kidney disease (3/12), pancreatitis (2/12), extrahepatic biliary tract obstruction (1/12), bacterial cholecystitis (1/12), hypercalcemia of unknown origin (1/12), suspected chronic pancreatitis (1/12) and neutrophilic and lymphoplasmacytic cholangitis (1/12).

3.2 | Clinical Signs

The clinical signs presented in the LGITL and LPE group are shown in Table 2. The median duration of clinical signs was 548 days (353–760; 24–1860) and 353 days (209–1095; 62–1860) for the LGITL and LPE group, respectively. No statistical differences were found between the two groups for any variable.

3.3 | Hematology and Serum Biochemistry

Serum biochemistry and CBC variables are available in Tables S2 and S3, respectively. No significant difference between the groups was identified.

TABLE 2 | Clinical signs of cats diagnosed with low-grade intestinal T-cell lymphoma (LGITL) and lymphoplasmacytic enteritis (LPE).

Clinical sign	LGITL ($n=15$)	LPE ($n=12$)	p[a]
Signs of GI disease	15 (100%)	10 (83%)	0.188
Lethargy	5 (33%)	5 (42%)	0.706
Hyporexia	8 (53%)	7 (58%)	1
Polyphagia	3 (20%)	1 (8%)	0.605
Weight loss	13 (87%)	11 (92%)	1
Vomiting	12 (80%)	10 (83%)	1
Constipation	0	1 (8%)	0.464
Small intestine diarrhea	7 (47%)	6 (50%)	1
Hematochezia	2 (13%)	1 (8%)	1
Melena	2 (13%)	1 (8%)	1
Colic diarrhea	2 (13%)	1 (8%)	1
Pyrexia	0	1 (8%)	0.464
Abnormal abdominal palpation	6 (40%)	7 (58%)	0.449
Painful	0	1 (8%)	0.464
Thickened intestinal loops	6 (40%)	4 (33%)	1
Dilated with liquid	0	2 (17%)	0.188

Note: All data are expressed as number of cats affected and proportion (%).
Abbreviation: GI, gastrointestinal.
[a]Reported P values are for either χ^2 test or Fisher's exact test (where expected values were <5 for at least 1 in the contingency table).

TABLE 3 | Reports of lesions during endoscopic stomach evaluation in cats with low-grade intestinal T-cell lymphoma (LGITL) and lymphoplasmacytic enteritis (LPE).

	LGITL	LPE
No macroscopic lesions	4/12 (33%)	2/10 (20%)
Presence of macroscopic lesions	8/12 (66%)	8/10 (80%)
Erosions (1–3/3)	4/8 (50%)	2/8 (25%)
1/3	3	2
3/3	1	/
Edema (1–3/3)	7/8 (88%)	6/8 (75%)
1/3	3	5
2/3	3	1
3/3	1	/
Hyperhemia (1–3/3)	2/8 (25%)	4/8 (50%)
1/3	1	3
2/3	1	1
Discoloration (1–3/3)	3/8 (38%)	/
1/3	3	
Mucosal hemorrhage (1–3/3)	1/8 (12%)	1/8 (12%)
1/3	/	1
2/3	1	/
Pyloro-duodenal reflux	2/8 (25%)	1/8 (12%)
Pylorus stenosis	3/12 (25%)	4/10 (40%)
Difficulties in pyloric catheterization	2/12 (17%)	3/10 (30%)

Note: Categorical data are expressed as number (%).

3.4 | Abdominal US Evaluation

All cats in the LGITL group and 11/12 cats in the LPE group had their US examination performed at the referral center. One cat in the LPE group had an abdominal US performed 2 weeks before referral by an experienced veterinarian in imaging diagnosis.

The retrospective review of assessed variables during abdominal US is detailed in Table S4. There were no statistical differences regarding the presence of abdominal effusion and the thickness, echogenicity, and structure of jejunal lymph nodes ($p>0.05$).

3.5 | Endoscopic Findings

Endoscopic reports were available for review in 13/15 cats in the LGITL group and 10/12 cats in the LPE group. All except one cat underwent gastric and duodenal endoscopic evaluation and sampling.

Stomach evaluation did not reveal abnormalities in 33% and 20% of cats diagnosed with LGITL and LPE, respectively. No macroscopic lesions were observed during duodenal evaluation in 56% and 33% of those same cats, respectively. The detailed results of reported lesions during upper GI endoscopy are reported in Tables 3 and 4. The pylorus could not be intubated in 25% of cats diagnosed with LGITL and 40% with LPE, presumably because of acquired pyloric narrowing. There was no statistical difference regarding the proportion of suspected pyloric stenosis in cats diagnosed with LGITL and LPE.

Concerning ileal sampling, in the LGITL group, blinded endoscopic biopsies were performed in 3 cats, while surgical biopsies were obtained in 12 cats. Unsuccessful ileo-colic junction catheterization was specifically reported in 1 out of the 3 LGITL cats that underwent endoscopic ileal biopsies. In the LPE group, ileum biopsies were obtained through endoscopy in 4 cats and surgically in 8 cats. Unsuccessful ileo-colic junction catheterization was reported in 2 out of the 4 LPE cats that had endoscopic ileal biopsies.

Endoscopic colonic mucosa sampling was performed in 3 cats in the LGITL group and in 5 cats in the LPE group. No macroscopic lesion was reported in the LGITL cats and in 3 cats in the LPE

TABLE 4 | Reports of lesions during endoscopic duodenal evaluation in cats with low-grade intestinal T-cell lymphoma (LGITL) and lymphoplasmacytic enteritis (LPE).

	LGITL	LPE
Median duodenal catheterism length (cm)	7/9 20 cm (10–30; 10–40)	5/6 20 cm (10–25; 10–30)
No macroscopic lesions	5/9 (56%)	2/6 (33%)
Presence of macroscopic lesions	4/9 (44%)	4/6 (67%)
Erosions (1–3/3)	/	/
Oedema (1–3/3)	/	2/4 (50%)
1/3		2
Hyperhemia (1–3/3)	1/4 (25%)	/
1/3	1	
Discoloration (1–3/3)	3/4 (75%)	1/4 (25%)
1/3	2	1
2/3	1	/
Mucosal hemorrhage (1–3/3)	/	/
Excessive granularity	3/4 (75%)	2/4 (50%)
Lymphatic dilation	/	1/4 (25%)

Note: Categorical data are expressed as number (%).

group. The two other LPE cats had mucosal hyperemia (2 cats; graded 1/3 and 2/3 respectively), erosions (1 cat; graded 2/3), and oedema (1 cat; graded 2/3).

3.6 | Histologic Findings

3.6.1 | LGITL Group

In the 15 cats diagnosed with LGITL, neoplastic infiltrate was observed in the stomach of 4/14 cats (29%), in the duodenum of 13/15 cats (87%), in the jejunum of all cats (15/15, 100%), in the ileum of 14/15 cats (93%), and in the colon of 2/3 cats. Of the 3 cats with presumed acquired pyloric narrowing and blinded duodenal biopsies, all of them had neoplastic infiltration of the duodenum on histopathologic evaluation. Involvement of all GI segments or only jejunum was not observed in any cats. The combination of affected segments was duodenum/jejunum/ileum in 7 cats, stomach/duodenum/jejunum/ileum in 4 cats, and jejunum/ileum or duodenum/jejunum/ileum/colon in 2 cats, respectively.

The median severity of histopathologic score (IQR; minimal and maximal values) for histologic lesions in the stomach, duodenum, jejunum, and ileum was 0 (0–1; 0–1), 2 (1.5–3; 0–3), 3 (2.5–3; 1–3) and 2 (2–3; 1–3), respectively, and was statistically different between the 4 GI segments ($p < 0.0001$). Paired comparisons showed that the severity score of the stomach was significantly lower compared to the duodenum, jejunum, and ileum ($p < 0{,}0001$ respectively). However, no significant difference was found between the other segments.

Interestingly, concomitant inflammatory lesions and neoplastic infiltrates were found in 9/15 cats (60%), mostly in the duodenum (6/9), followed by the stomach (4/9), jejunum (3/9) and colon (2/9).

3.6.2 | LPE Group

In the 12 cats diagnosed with LPE, inflammatory infiltrates were observed in the stomach of 4/12 cats (33%), in the duodenum and jejunum of all cats (12/12, 100%, respectively), in the ileum of 11/12 cats (92%), and in the colon of 3/4 cats. The combination of affected segments was duodenum/jejunum/ileum in 6 cats, stomach/duodenum/jejunum/ileum in 3 cats, duodenum/jejunum/ileum in 2 cats, and all GI segments in one cat. No cats had jejunum or jejunum/ileum solely affected.

3.7 | Diagnosis Accuracy Between Endoscopic-Guided and Surgical Biopsies

As the jejunum was always involved in both inflammatory and neoplastic lesions, jejunal surgical biopsies led to a diagnosis of LGITL or LPE in all cats.

Ileal surgical or endoscopic-guided biopsies were diagnostic in 93% of LGITL cases and 92% of LPE cats. Upper GI endoscopy with stomach and duodenal biopsy evaluation was diagnostic in 87% (13/15) of LGITL cats. Thus, misdiagnosis was possible in 13% (2/15) of confirmed LGITL cases if only upper GI endoscopy had been performed. In the 3 cats with endoscopic-guided biopsies of the stomach, duodenum, and ileum, LGITL was diagnosed in all cases. Combining upper GI endoscopy with ileal biopsies (whether obtained endoscopically guided [3/15] or surgically [12/15]) led to a diagnosis of LGITL in all cats.

4 | Discussion

This prospective study aimed to assess the distribution of GI lesions in a cohort of cats ultimately diagnosed with LPE and LGITL, using both endoscopic and surgical biopsies during the same procedure. It further aimed to analyze lesion severity and associated inflammation in LGITL cats, and to describe clinical presentation, laboratory and US findings, endoscopic lesions and histopathological data observed in this cohort. The results showed that LGITL exhibits a heterogeneous distribution within the digestive tract, with the jejunum involved in all cats, followed by the ileum and duodenum. Even though surgical biopsies of the jejunum appear to be the best sampling method to obtain a definitive diagnosis, upper and lower GI sampling provided a precise diagnosis in all cases in our study. Therefore, when all segments can be accessed, upper and lower GI endoscopy could be considered an accurate minimally invasive diagnostic method.

This study agrees with the reported distribution of neoplastic lesions in LGITL cases [2, 7, 10–13]. The jejunum was always involved, followed closely by ileal infiltration in 93% of cats and duodenal involvement in 87% of cats. The stomach was infrequently affected, as previously reported. However, because immunohistochemistry was not mandatory for stomach samples, it is possible that some emerging neoplastic lesions might have been missed. Colonic infiltration with neoplastic cells could not be conclusively assessed due to the lack of systematic colonic histopathology; however, it was observed in two out of three LGITL in which colonic samples were obtained. Cats affected by LPE had consistent jejunal and duodenal involvement, followed by ileal infiltration in 92% of cases. All cats had at least two GI segments affected by the given disease, challenging the belief that jejunal biopsies are always necessary to reach a diagnosis of LGITL. This also highlights that upper and lower endoscopic-guided biopsies might be reliable for diagnosing these cases when both duodenal and ileal biopsies are feasible. As previously mentioned, clonality was not mandatory in this study, as recent publications confirm a high percentage of clonality in LPE cases probably due to emerging clones in a *continuum* background [2, 6].

As previously reported in the feline literature, more than a half of LGITL cases showed concomitant inflammatory lesions [3, 6, 24]. Previous studies have already reported the concomitant presence of segmental inflammatory lesions and neoplastic lymphocyte infiltration of varying grades [25]. Recent articles reveal that LGITL lesions showed an apical-to-basal gradient in nearly half of the cases, suggesting that chronic antigenic stimulation could lead to the emergence of neoplastic cells. More precisely, it is suspected that chronic stimulation would trigger the expansion of T-cell lymphocytes, cytokine production, and the recruitment of other immune and inflammatory cells, and thus T-cell proliferation. Additionally, genetic events like JAK/STAT pathway dysregulation might also contribute to the emergence of T-cell clones and their expansion [24]. Although our study was not designed to confirm the suspected *continuum* between the two diseases, the concomitant presence of inflammatory and neoplastic lesions in a large proportion of cases in the same area can support this hypothesis [3, 6, 24].

The evaluation of the median severity score of histologic lesions in LGITL cases revealed that the stomach had a significantly lower score compared to the duodenum, jejunum, and ileum. Although a statistical difference was not reached, higher scores were observed in the jejunum. If the small intestine, particularly the jejunum, could be the hypothetical emerging site of neoplastic lesions, higher lesion scores would be expected in this segment, as reported in our study. However, the number of cats might have been too small to reach significance, and only one certified pathologist assessed this subjective score, which could have led to bias.

This study also aimed to compare the diagnostic accuracy of surgical versus endoscopically assisted intestinal biopsies. Since the jejunum was consistently affected in LGITL cases, jejunal surgical biopsies led to the correct diagnosis in all cases. However, this procedure can be challenging in elderly cats with comorbidities, and surgical complications occur in 0%–15% of cases in the literature [20, 21]. Moreover, full-thickness biopsies are not mandatory in LGITL as lesions systematically emerge from the epithelium to the lamina propria [6, 24]. Less invasive and quicker procedures such as needle-core biopsies seem to limit the associated complications and warrant further studies [26].

Endoscopy showed good to excellent diagnostic performance in our study. When duodenal and ileal biopsies were obtained, and thus if upper and lower GI biopsies were performed during the same anesthetic time, a diagnosis of LGITL was established, consistent with the findings from jejunal surgical biopsies. However, when only upper or lower GI biopsies were obtained, a misdiagnosis was possible in 13% and 7% of cases, respectively. Previous studies have reported a moderate to poor agreement between upper and lower GI biopsies [17, 18]. More precisely, LGITL was diagnosed only by duodenal biopsies in 17%–39% of cases and only by ileal biopsies in 2%–44% of cats, depending on the study and the histopathologic, immunohistochemistry, and clonality features used for diagnosis. Therefore, based on our results, bidirectional GI endoscopic evaluation of cats presented for suspected LGITL or LPE might allow an accurate diagnostic if duodenal and ileal segments can be reached.

It is noteworthy that most cats in our study underwent surgical ileal biopsies rather than endoscopic-guided biopsies. Anesthesia and endoscopic duration, along with difficulties in ileal catheterization, were the most common reasons for performing ileal surgical biopsies concurrently with jejunal biopsies. To our knowledge, the comparative diagnostic accuracy of endoscopic versus surgical ileal biopsies in cats remains unknown. However, given that the median length of ileum in cats is approximately 2.5 cm, we assumed that if ileal catheterization is achievable, the same segment could be reached in endoscopy as with surgery [27].

As suggested in the literature, LGITL lacks specific macroscopic lesions on endoscopic evaluation [13, 28]. In our study, common findings in the stomach and duodenum include granularity, erosions, mucosal discoloration, or edema. Surprisingly, 33% and 56% of LGITL cats had no visible lesions on endoscopy of the stomach and duodenum, respectively. In contrast, one study reports duodenal lesions in all LGITL and LPE cases, mostly mild to moderate [13]. Given the conflicting literature, it is believed that the absence of lesions on endoscopic evaluation should never rule out LGITL from the differential diagnosis in these patients. Moreover, presumed acquired pyloric narrowing was reported in 25% of LGITL cases in our study, higher than previously reported [19].

The use of magnifying endoscopy combined with narrow band imaging in human medicine has revealed characteristic features in intestinal T-cell lymphoma, colorectal polyps, and gastric mucosal associated lymphoid tissue lymphoma cases [29–31]. Whether this technique could assist in selecting the best site to sample or help to differentiate LGITL from LPE in cats remains unknown, but this promising technique warrants further studies.

Most of the cats included in this study were mature to senior [1–4, 8–10]. The youngest cat diagnosed with LGITL was 8 years old, confirming that LGITL is not a disease of young adults. No significant difference was found regarding the epidemiologic,

clinical, and laboratory findings between LGITL and LPE cats, highlighting the challenge to differentiate the two diseases. These results differ from a previous study involving 44 adult cats, where a longer duration of clinical signs, male sex, polyphagia, and hypocobalaminemia are significantly more frequent in LGITL cats compared to LPE [1]. In our study, cats with polyphagia (20% versus 8%) and hypocobalaminemia (57% versus 33%) seemed to be numerically higher in LGITL cases, and LGITL cats appeared to have a longer duration of clinical signs in our cohort (548 versus 353 days) compared to LPE. Additionally, abdominal US findings did not differ significantly between the two diseases, unlike in the same study, where abdominal effusion and lymph node structure and echogenicity are discriminating factors [1]. However, in this study, even if statistical significance was not reached, abdominal effusion was more frequently recorded in LGITL cases (60% versus 33%). Due to the longer duration of coupled endoscopy and surgery compared with endoscopy alone, only a small number of cats were included in this study. Larger studies are needed to confirm the accuracy of these indicators to differentiate LGITL and LPE.

This study has several limitations. First, only a small number of cats from a referral center sample were included, which might have resulted in insufficient statistical power. Second, only one pathologist blindly reviewed all the histopathologic and immunohistochemistry results, which might have led to some bias.

Third, although this study was prospectively conducted, sampling was not standardized across all cases. Specifically, the jejunum and ileum biopsy sites were selected by different surgeons, which could have affected diagnostic accuracy, despite the selection being based on US findings and careful palpation of the GI tract during surgery. Similarly, most ileal biopsies were performed surgically, and it remains unknown whether endoscopic-guided ileal biopsies would provide the same diagnostic accuracy as surgical samples. Additionally, as LGITL is known to primarily affect the small intestine, the colon was not specifically targeted in this study. Colonic biopsies were not systematically performed due to the prolonged duration of endoscopic procedures and the high rate of complications associated with full-thickness surgical biopsies of the colon.

Finally, another limitation is the lack of systematic immunohistochemical evaluation in some segments, particularly the stomach, which might have led to undetected neoplastic infiltration. While CD3 and CD20 staining were consistently performed in commonly affected segments (jejunum, duodenum, ileum), Ki67 evaluation was not mandatory and was mainly performed in ambiguous cases to help distinguish LGITL from LPE. However, since no LGITL cats in this study showed neoplastic lymphocyte infiltration in a single segment, it is unlikely that jejunal, duodenal, and ileal neoplastic infiltration were overlooked in the same cat. Therefore, it seems unlikely that some LPE cases might have been misclassified.

5 | Conclusions

To conclude, our results suggest that sampling both duodenum and ileum allowed an accurate diagnosis of LGITL in all cases. Misdiagnosis can happen if only duodenal or ileal biopsies are performed in 13% and 7% of cases, respectively. Therefore, endoscopic-guided biopsies of both the upper and lower GI tract can be considered a first-line option to establish diagnosis, despite the lack of distinctive macroscopic lesions. LGITL showed a heterogeneous distribution of lesions, with the jejunum systematically infiltrated, closely followed by ileal and duodenal infiltration. Concurrent inflammatory lesions were found in 60% of LGITL cases, further supporting the possible existence of a recently suggested *continuum* between those two entities.

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Approved by the Scientific Ethics Committee ENVA COMERC no 2017-05-09. Authors declare human ethics approval was not needed.

Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.